2023

COMPUTER SCIENCE AND ENGINEERING

Paper : CSEL-1016

(Real Time System)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer ***question nos.*** **1**, **2** and ***any four*** questions from the rest.

1. Answer ***any five*** from the following (comment critically) : 2×5

   (a) Cyclic schedulers are more proficient than the table-driven schedulers.

   (b) Finding an optimal schedule for a set of independent periodic hard real-time systems without any resource-sharing constraints under a static-priority scheme is an NP-complete problem.

   (c) The worst-case response time of a task occurs when it is out of phase with other higher-priority tasks.

   (d) RMA should be preferred over the time-sliced round-robin algorithm for scheduling a set of soft real-time tasks on a uni-processor system.

   (e) The primary priority ceiling protocol is immune to deadlock.

   (f) When periodic real-time tasks are scheduled using RMA, specific critical resources share priority inheritance protocol, and some might suffer unbounded priority inversion.

   (g) Priority ceiling protocol is free from deadlock and unbounded priority inversions.

   (h) The traditional 2PL protocol is not suitable for real-time databases.

2. Answer ***any five*** from the following : 4×5

   (a) Critically comment on the desirable characteristics of a real-time system.

   (b) What is a critical instant? What are its implications for scheduling fixed-priority tasks for RTS?

   (c) Critically comment on necessary and sufficient schedulability tests for a fixed priority system. What are the practical factors whose effect needs to be considered for such a schedulability test?

   (d) What do you mean by the 'fail-safe' state of the system? What is the statement's implication: '*Safety-critical real-time systems do not have a fail-safe state*'?

   (e) What do you mean by fault tolerance? Compare and contrast hardware and software fault tolerance.

   (f) Critically comment on real-time communication systems.

   (g) What protocols and standards are supported by the IoT system to enable seamless integration with existing infrastructure and legacy systems?

   (h) Give an example, where some jobs cannot meet their deadlines if the tasks are scheduled dynamically on the processors according to the 'Least Slack Time' algorithm.

Please Turn Over

3. (a) Prove that the Rate monotonic scheduling algorithm is an optimal static priority algorithm.

(b) What do you mean by jitter associated with a periodic task? How are the jitters caused? How can these be overcome?

(c) (i) Mention the constraints to determine the frame size for a given set of periodic tasks.

(ii) Comment on Tick Scheduling. 3+3+4

4. The periodic tasks (3, 1), (4, 2), and (6, 1) are scheduled according to the rate monotonic algorithm.

(a) Draw the time-demand functions of the tasks.

(b) Are the tasks schedulable? Why or why not?

(c) Can this graph be used to determine whether the tasks are schedulable according to an arbitrary priority-driven algorithm? Explain your answer. 3+3+4

5. (a) Justify the requirement of an acceptance test to schedule sporadic jobs along with periodic and aperiodic jobs.

(b) What is a deferrable server, and how is it different from a polled server? Explain the operation of a deferrable server with a suitable example.

(c) How the sporadic server is different from the deferrable server? What are the consumption and replenishment rules of sporadic servers? Explain with an example. 2+4+4

6. (a) For the following set of jobs with associated parameters, find the job schedule using the priority ceiling protocol.

Job	Release time	Execution time	Priority	Critical Section
J1	7	3	1	[Shaded;1]
J2	5	3	2	[black;1]
J3	4	2	3	No resource
J4	2	6	4	[Shaded;4[Black;1.5]]
J5	0	6	5	[Black;4]

(b) Explain how PCP is able to avoid deadlock, unbounded priority inversions, and chain blocking. 7+3

7. (a) What is priority inversion? When does it occur?

(b) Two options for avoiding priority inversion are the basic priority inheritance protocol and the basic priority ceiling protocol. Discuss how these protocols differ regarding scheduling, resource allocation, and priority inheritance rules.

(c) Can the basic priority ceiling protocol be used with dynamic priority scheduling? Would it be a good idea to do so? Explain your answer. 2+6+2

8. (a) Suppose we have two periodic processes P1 and P2, which measure pressure and temperature, respectively. The absolute validity interval of both of these parameters is 100 ms. The relative validity interval of a temperature-pressure pair is 50 ms. What is the maximum period of P1 and P2 that ensures that the database system always has a valid temperature-pressure pair reading?

(b) Why are the concurrency control protocols used in conventional database systems useless for real-time database systems? What is the information that real-time database schedulers can use?

(c) How does the IoT system address the challenges of power consumption and energy efficiency to maximize the lifespan of battery-powered devices? 4+3+3